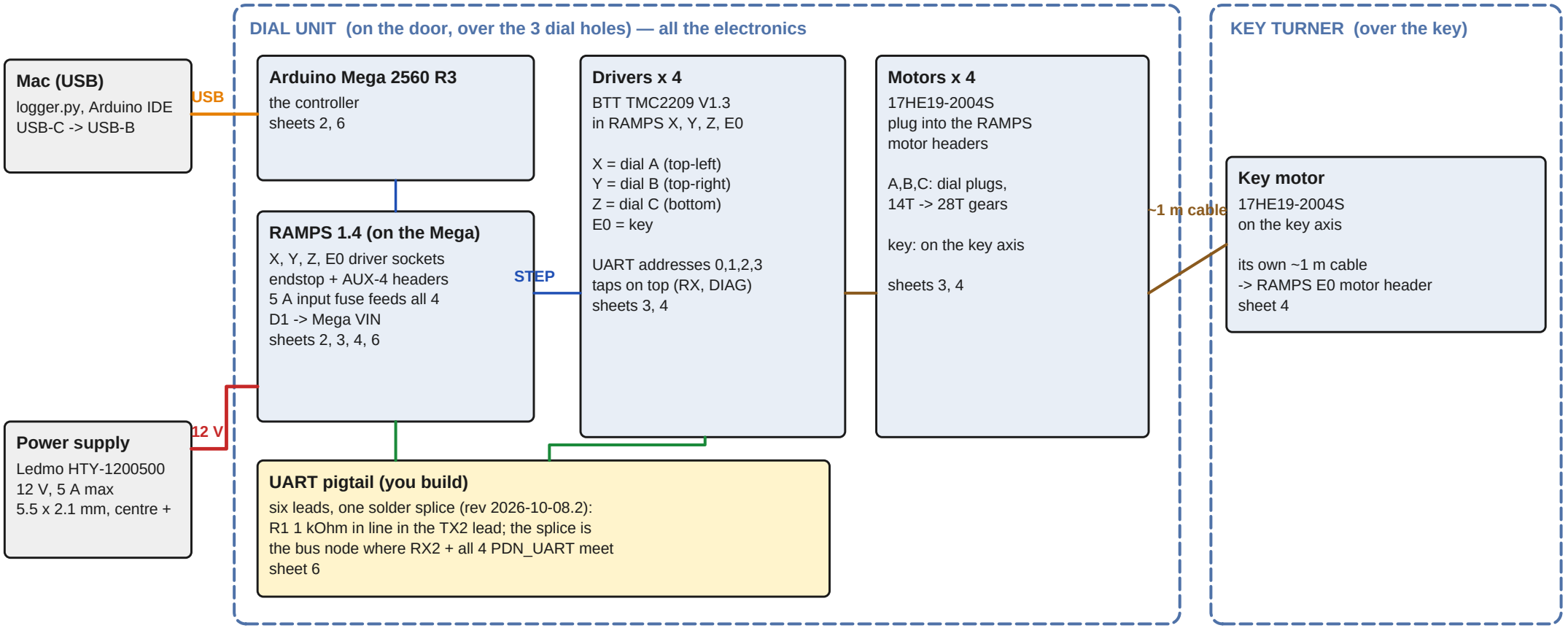
Overview

What connects to what. Each block is drawn in full on the sheet named in it.



Sheets

1 Overview (this sheet)
2 Power: 12 V, 5 V, GND
3 Drivers A and B on the RAMPS
4 Driver C and the key driver (E0)
5 Driver taps (DIAG, RX), orientation, key cable
6 Mega + RAMPS headers, UART pigtail
7 Pin map, jumpers, currents, parts

Rules

Make every connection with power off (USB unplugged, 12 V off).
Never plug or unplug a motor with 12 V on (the key motor cable included).
Power up: USB first, then 12 V. Power down: 12 V first.
Never fit an MS3 jumper: on the V1.3 that pin is the UART line.
Emergency stop: pull the 12 V plug. `!` aborts a move.
VERIFY = not yet checked on the bench (control/bringup.md).
Rev 2026-10-08.1: key driver in E0. Rev 2026-10-08.2: UART pigtail, no hub.
If this disagrees with control/wiring.md, wiring.md wins.

+12 V · +5 V (logic, VIO) · GND · UART bus
STEP / DIR / EN, button · DIAG (stall) · motor coils · jumper / tie / spare
bought module · board you build · already on the RAMPS / Mega: nothing to buy

Fichet safe robot — wiring schematic	Sheet 1 of 7
Overview	
2026-10-08 · revision 2026-10-17.1 · from control/wiring.md	

Power: 12 V, 5 V, GND

12 V goes from the PSU straight into the RAMPS '5A' input (its own 5 A fuse feeds all four drivers). 5 V logic comes from the Mega.

PSU
Ledmo HTY-1200500
12 V, 5 A max
5.5 x 2.1, centre +

DC jack
5.5 x 2.1 ->
screw terminal
VERIFY + / -

20 AWG red -> RAMPS '5A' +

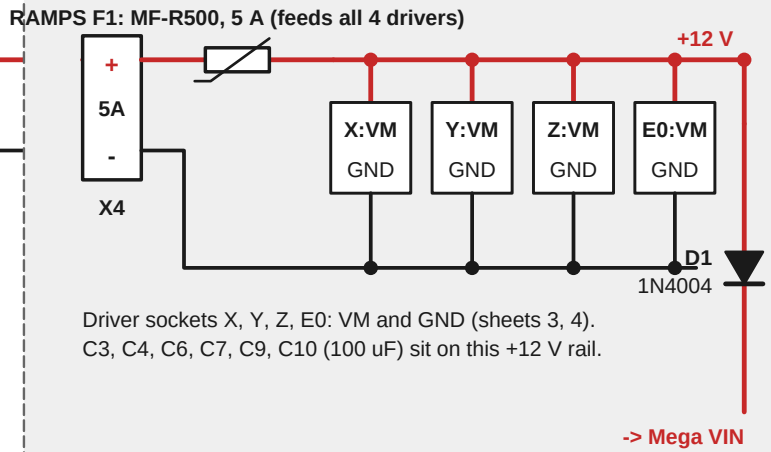
20 AWG black -> RAMPS '5A' -

Revision 2026-10-08.1:
12 V goes straight from the PSU adapter into the RAMPS '5A' terminal; RAMPS's own 5 A fuse feeds all four drivers. Nothing else carries 12 V (rev 2026-10-08.2: no hub board).

Not the Mega's own barrel jack: that feeds only the Mega (VIN). The drivers get 12 V only through the RAMPS '5A' input; D1 passes power from there to the Mega, never back.

RAMPS 1.4 - 12 V (already on the board)

from the RAMPS 1.4 KiCad netlist (control/wiring.md log, 2026-10-07)



Driver sockets X, Y, Z, E0: VM and GND (sheets 3, 4).
C3, C4, C6, C7, C9, C10 (100 uF) sit on this +12 V rail.

'11A' input (heated bed, through RAMPS F2): not used.
Fuses: PSU 5 A max; RAMPS F1 5 A (all 4 drivers + Mega VIN).
Revision 2026-10-08.1: no separate key-branch fuse.

5 V logic (VIO) - from the Mega

VIO powers only each driver's I/O pins; its logic runs from its own VM (12 V)

Arduino Mega 2560
5 V from USB, or its regulator (from VIN)

+5 V rail (RAMPS VCC)

- VIO of X, Y, Z, E0 — sheets 3, 4
- MS1 / MS2 jumpers — sheets 3, 4
- endstop headers' '+' pins — sheet 6

All four drivers get VIO from the same 5 V rail in their sockets (revision 2026-10-08.1): no 5 V over a cable any more.
Power order: USB first (Mega + every VIO), then 12 V. Off: 12 V first. The firmware waits until all 4 drivers answer.

+12 V · +5 V (logic, VIO) · GND · UART bus
STEP / DIR / EN, button · DIAG (stall) · motor coils · jumper / tie / spare
bought module · board you build · already on the RAMPS / Mega: nothing to buy

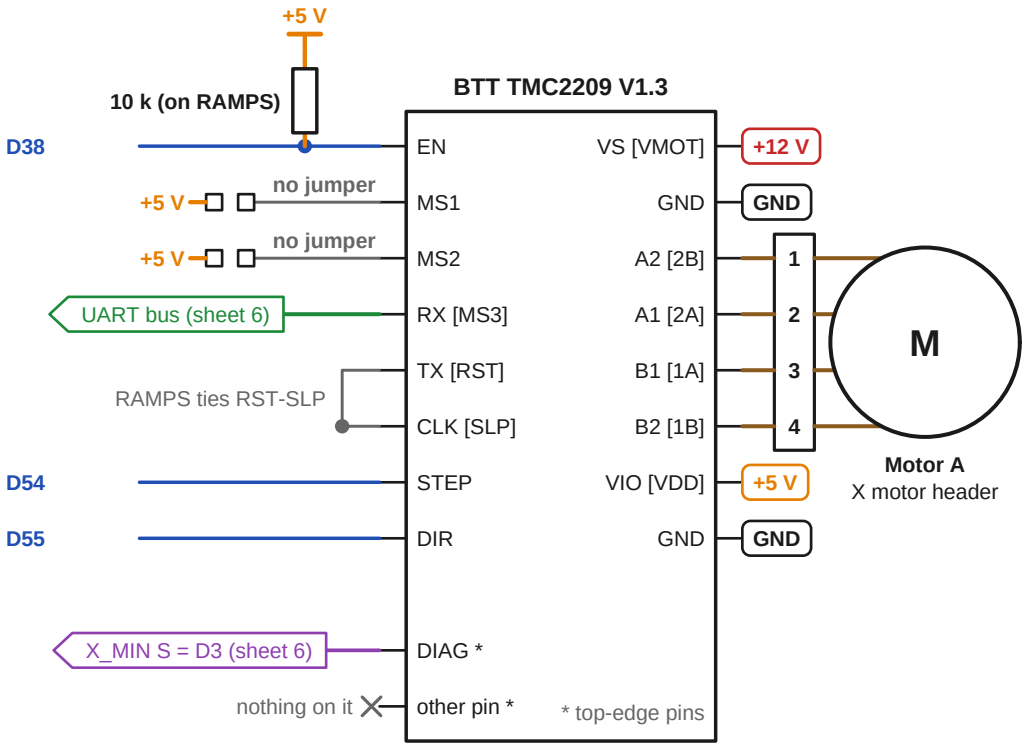
Fichet safe robot — wiring schematic	Sheet 2 of 7
Power: 12 V, 5 V, GND	
2026-10-08 · revision 2026-10-17.1 · from control/wiring.md	

Drivers A and B on the RAMPS

[ ] = the RAMPS socket name of that pin. All connections to the left are on the RAMPS or by jumper.

Socket X - dial A (top-left)

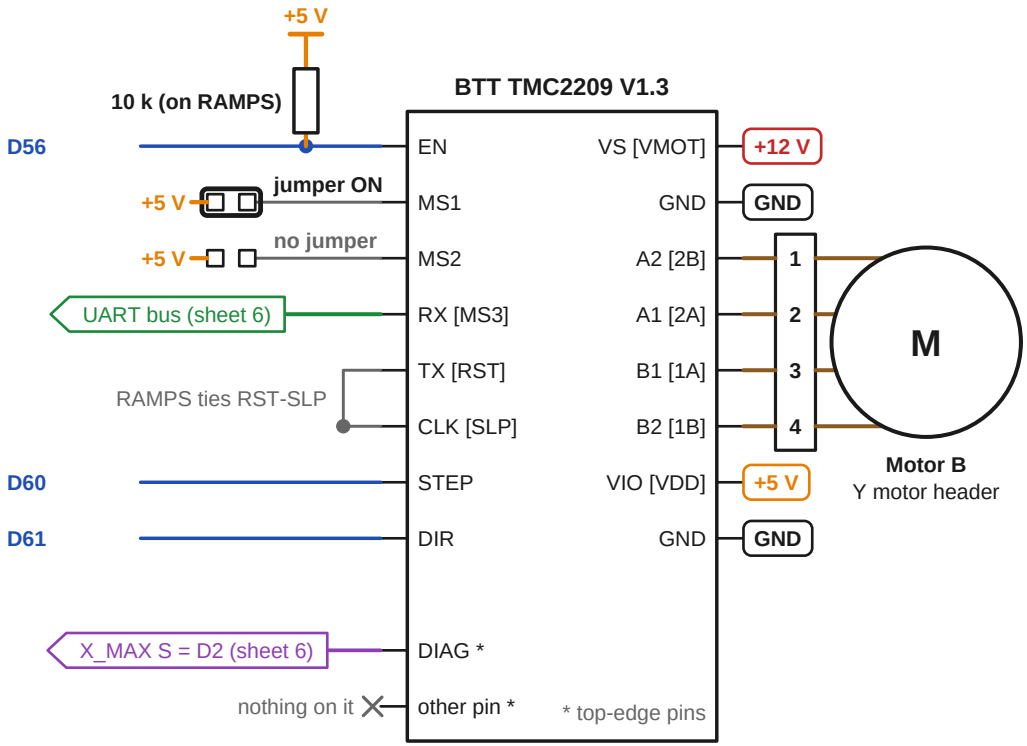
UART address 0 - BTT TMC2209 V1.3 (EN-end pins cut underneath)



Jumpers under socket X: MS1 off, MS2 off, MS3 off (never fit MS3)
UART lead: clipped onto the top of the driver's RX pin (sheet 5).
STEP D54 - DIR D55 - EN D38. The RAMPS pull-up keeps the driver off until the firmware pulls EN low.
DIAG: F-F jumper, top of DIAG pin (sheet 5) -> X_MIN S: D3, an interrupt.
Motor 17HE19-2004S: header pins 1-2 = one coil, 3-4 = the other.

Socket Y - dial B (top-right)

UART address 1 - BTT TMC2209 V1.3 (EN-end pins cut underneath)



Jumpers under socket Y: MS1 ON, MS2 off, MS3 off (never fit MS3)
UART lead: clipped onto the top of the driver's RX pin (sheet 5).
STEP D60 - DIR D61 - EN D56. The RAMPS pull-up keeps the driver off until the firmware pulls EN low.
DIAG: F-F jumper, top of DIAG pin (sheet 5) -> X_MAX S: D2, an interrupt.
Motor 17HE19-2004S: header pins 1-2 = one coil, 3-4 = the other.

[ ] = the RAMPS / StepStick socket name of that pin. VS/GND/motor row in the socket row nearer the motor header (sheet 5): a reversed driver is destroyed.
The BTT V1.3's RX pin is PDN_UART, in the socket's MS3 position. TX: R10 not fitted (RX-TX measured open), so TX connects to nothing. Don't bridge R10.
CLK: the RAMPS ties the socket's RST and SLP pins (= TX and CLK on the V1.3). With R10 open, CLK sees only its 20 k pull-down -> internal clock.
Motor currents are set over UART: 1.0 A RMS, hold 0.5 A. If a motor buzzes instead of turning, swap the middle two wires in its plug.

+12 V · +5 V (logic, VIO) · GND · UART bus
STEP / DIR / EN, button · DIAG (stall) · motor coils · jumper / tie / spare
bought module · board you build · already on the RAMPS / Mega: nothing to buy

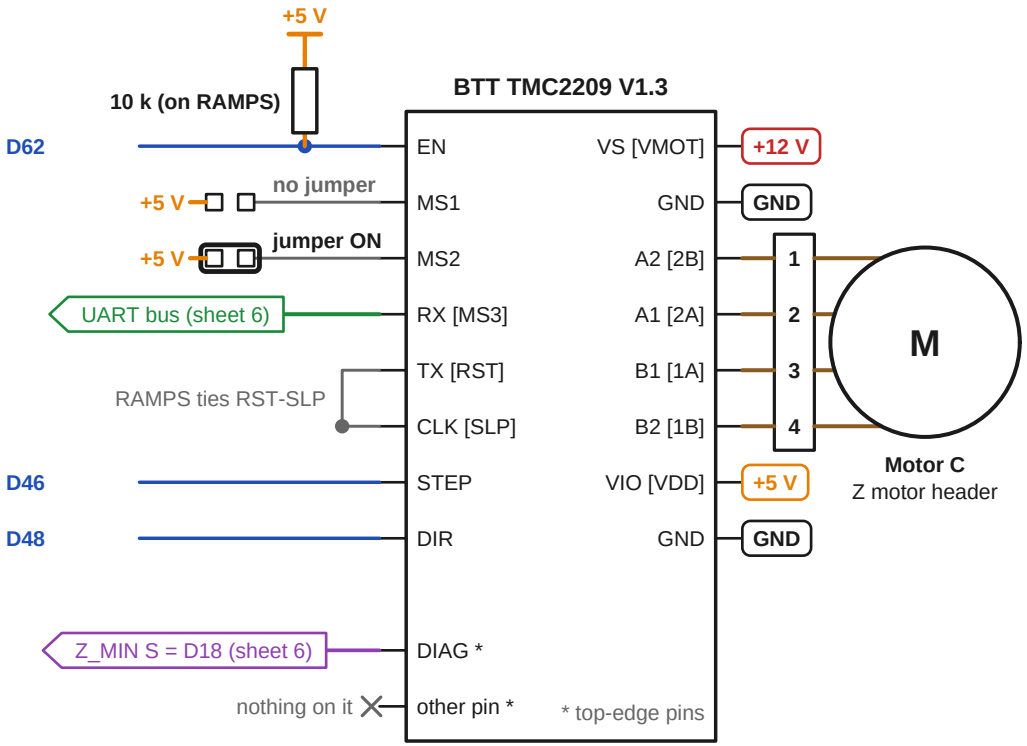
Fichet safe robot — wiring schematic	Sheet 3 of 7
Drivers A and B on the RAMPS	
2026-10-08 · revision 2026-10-17.1 · from control/wiring.md	

Dial driver C and the key driver (E0)

The key driver is an ordinary driver in the E0 socket (revision 2026-10-08.1).

Socket Z - dial C (bottom)

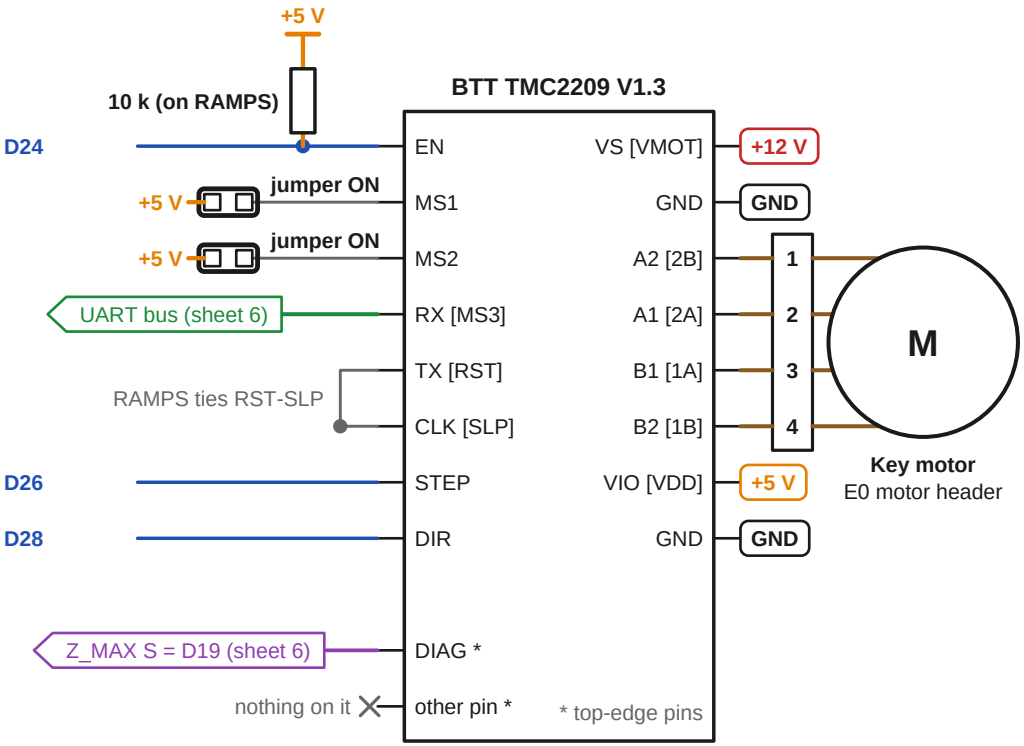
UART address 2 - BTT TMC2209 V1.3 (EN-end pins cut underneath)



Jumpers under socket Z: MS1 off, MS2 ON, MS3 off (never fit MS3)
UART lead: clipped onto the top of the driver's RX pin (sheet 5).
STEP D46 - DIR D48 - EN D62. The RAMPS pull-up keeps the driver off until the firmware pulls EN low.
DIAG: F-F jumper, top of DIAG pin (sheet 5) -> Z_MIN S: D18, an interrupt.
Motor 17HE19-2004S: header pins 1-2 = one coil, 3-4 = the other.

Socket E0 - the key turner

UART address 3 - BTT TMC2209 V1.3 (EN-end pins cut underneath)



Jumpers under socket E0: MS1 ON, MS2 ON, MS3 off (never fit MS3)
UART lead: clipped onto the top of the driver's RX pin (sheet 5).
STEP D26 - DIR D28 - EN D24. The RAMPS pull-up keeps the driver off until the firmware pulls EN low.
DIAG: F-F jumper, top of DIAG pin (sheet 5) -> Z_MAX S: D19, an interrupt.
Motor 17HE19-2004S: header pins 1-2 = one coil, 3-4 = the other.

Revision 2026-10-08.1: the key driver is an ordinary driver in the RAMPS E0 socket, like the dials.
E0 address 3 = MS1 + MS2 jumpers ON. The key's STEP/DIR/EN are D26/D28/D24 (Marlin pins_RAMPS.h E0_STEP/DIR/ENABLE).
The key motor's own ~1 m cable plugs into the E0 motor header. Use it as supplied; shorten it only if the key's StallGuard reads too dull (bench stage 5).
Key current: 0.6 A RMS to start, hold 0.3 A. The key's UART (stage 1) and DIAG (stage 3) are bench checks like the dials'.

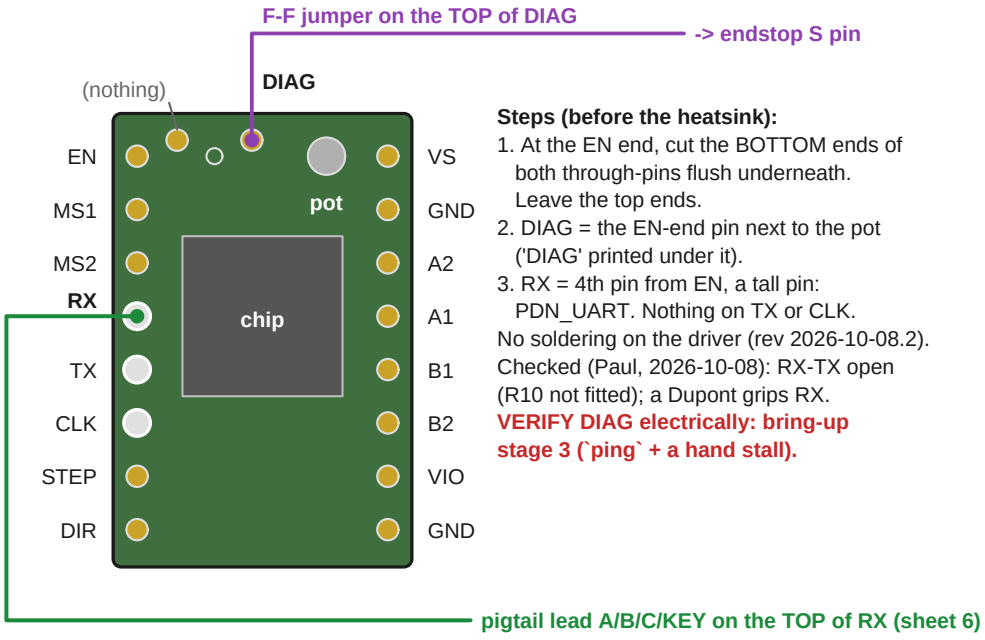
+12 V · +5 V (logic, VIO) · GND · UART bus
STEP / DIR / EN, button · DIAG (stall) · motor coils · jumper / tie / spare
bought module · board you build · already on the RAMPS / Mega: nothing to buy

Fichet safe robot — wiring schematic	Sheet 4 of 7
Dial driver C and the key driver (E0)	
2026-10-08 · revision 2026-10-17.1 · from control/wiring.md	

Driver taps (DIAG, RX), orientation; the key motor cable

Driver taps: DIAG and RX, from the top (all four drivers)

BTT TMC2209 V1.3 seen from above - not to scale - Paul's drivers, 2026-10-08



Steps (before the heatsink):

1. At the EN end, cut the BOTTOM ends of both through-pins flush underneath. Leave the top ends.
2. DIAG = the EN-end pin next to the pot ('DIAG' printed under it).
3. RX = 4th pin from EN, a tall pin: PDN_UART. Nothing on TX or CLK.

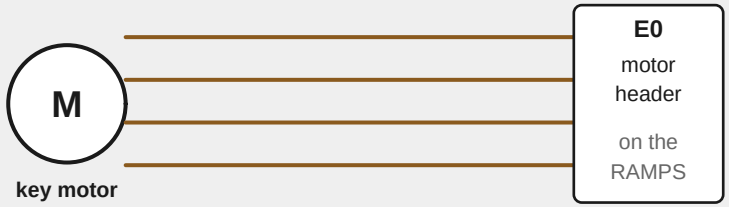
No soldering on the driver (rev 2026-10-08.2).
Checked (Paul, 2026-10-08): RX-TX open (R10 not fitted); a Dupont grips RX.
VERIFY DIAG electrically: bring-up stage 3 (`ping` + a hand stall).

Key motor cable (the only wire between the units)

revision 2026-10-08.1

Key turner -> dial unit

the motor's own ~1 m cable, 4 wires + connector



Plugs into the E0 motor header like a dial motor.
Use the supplied ~1 m cable as it is; shorten it only if the key's StallGuard reads too dull to find the stop (bench stage 5).
Never plug or unplug it with 12 V on.
If the motor buzzes instead of turning, swap its middle two wires.

Revision 2026-10-08.1 replaced the 6-conductor inter-unit signal cable and the remote driver board with this one motor cable.

Which way the driver goes (match the EN / GND silkscreen labels)

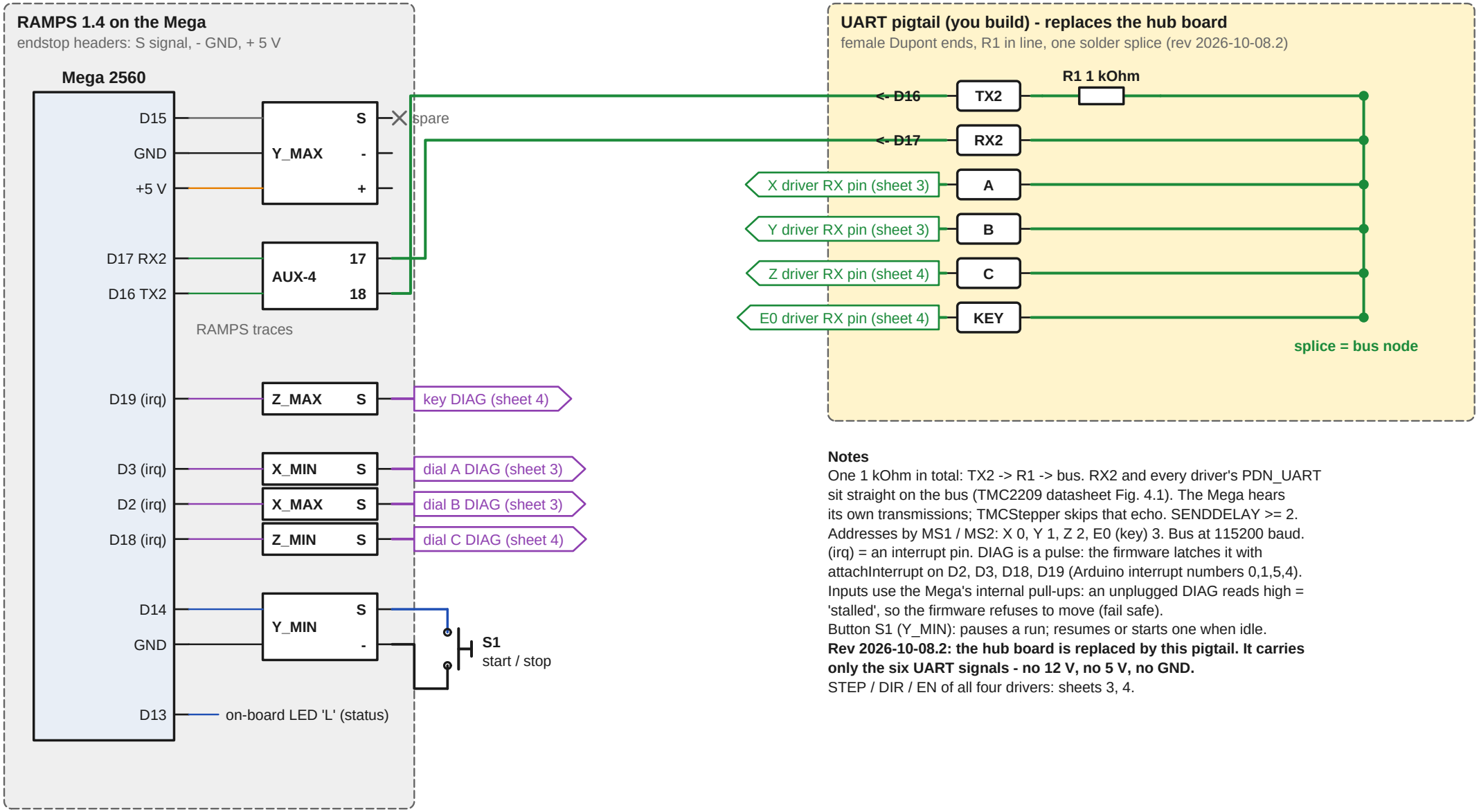
Driver EN and GND pins at the small EN / GND labels on the RAMPS silkscreen (no square pad on top).
The driver's VS / GND / motor row goes in the socket row NEARER that axis's 4-pin motor header; EN sits directly across from VS (RAMPS 1.4 KiCad: EN is the square pad underneath).
Cross-check if the labels are unclear (power off, no driver, continuity):
'5A' - to the motor-side row: 2 holes beep (GND, one at each end).
'5A' + to the end hole beside a GND: beeps through F1 = VS.
A reversed driver is destroyed. Manual 2.4 step 2.
MS3 jumper OFF on every socket (the MS3 position is RX = UART).

+12 V · +5 V (logic, VIO) · GND · UART bus
STEP / DIR / EN, button · DIAG (stall) · motor coils · jumper / tie / spare
bought module · board you build · already on the RAMPS / Mega: nothing to buy

Fichet safe robot — wiring schematic	Sheet 5 of 7
Driver taps (DIAG, RX), orientation; the key motor cable	
2026-10-08 · revision 2026-10-17.1 · from control/wiring.md	

Mega + RAMPS headers, UART pigtail

Thin lines = RAMPS traces. Thick lines = your jumper wires (F-F, 10-20 cm) and the UART pigtail.



Notes

One 1 kOhm in total: TX2 -> R1 -> bus. RX2 and every driver's PDN_UART sit straight on the bus (TMC2209 datasheet Fig. 4.1). The Mega hears its own transmissions; TMCStepper skips that echo. SENDDELAY >= 2.
Addresses by MS1 / MS2: X 0, Y 1, Z 2, E0 (key) 3. Bus at 115200 baud.
(irq) = an interrupt pin. DIAG is a pulse: the firmware latches it with attachInterrupt on D2, D3, D18, D19 (Arduino interrupt numbers 0,1,5,4).
Inputs use the Mega's internal pull-ups: an unplugged DIAG reads high = 'stalled', so the firmware refuses to move (fail safe).
Button S1 (Y_MIN): pauses a run; resumes or starts one when idle.
Rev 2026-10-08.2: the hub board is replaced by this pigtail. It carries only the six UART signals - no 12 V, no 5 V, no GND.
STEP / DIR / EN of all four drivers: sheets 3, 4.

+12 V · +5 V (logic, VIO) · GND · UART bus
STEP / DIR / EN, button · DIAG (stall) · motor coils · jumper / tie / spare
bought module · board you build · already on the RAMPS / Mega: nothing to buy

Fichet safe robot — wiring schematic	Sheet 6 of 7
Mega + RAMPS headers, UART pigtail	
2026-10-08 · revision 2026-10-17.1 · from control/wiring.md	

Pin map, jumpers, currents, parts

Pin map (control/wiring.md 1; Marlin pins_RAMPS.h) - revision 2026-10-08.2

Function	Mega pin	Where it connects
Dial A STEP / DIR / EN	D54 / D55 / D38	RAMPS X socket
Dial B STEP / DIR / EN	D60 / D61 / D56	RAMPS Y socket
Dial C STEP / DIR / EN	D46 / D48 / D62	RAMPS Z socket
Key STEP / DIR / EN	D26 / D28 / D24	RAMPS E0 socket
Dial A DIAG	D3 (interrupt)	X_MIN header, S
Dial B DIAG	D2 (interrupt)	X_MAX header, S
Dial C DIAG	D18 (interrupt)	Z_MIN header, S
Key DIAG	D19 (interrupt)	Z_MAX header, S (E0 driver DIAG pin)
UART TX2	D16	AUX-4 pin 18 -> pigtail TX2 (R1)
UART RX2	D17	AUX-4 pin 17 -> pigtail RX2
Drivers' PDN_UART	-	RX pins X,Y,Z,E0 -> pigtail A,B,C,KEY
Start / stop button	D14 (pull-up)	Y_MIN header, S and -
Status LED	D13	on-board LED 'L'
Spare	D15 / D23	Y_MAX S / AUX-4 16
USB serial (the log)	D0 / D1	USB: keep free

Power order (8)

1. Make every connection with power off (USB unplugged, 12 V off).
2. Before the first power-up: every driver's VREF pot to minimum.
3. Power up: USB first, then 12 V. 4. Power down: 12 V first, then USB.
5. Never plug or unplug a motor with 12 V on (the key motor cable too).
6. Emergency stop: pull the 12 V plug.

VERIFY on the bench (control/bringup.md)

Driver orientation in each socket (manual 2.4 step 2); all four answer `ping` (stage 1).
DIAG is the pin next to the pot (stage 3). The key answers at address 3 (E0).
The key's StallGuard through the 1 m motor cable (stage 5); shorten if dull.
Multimeter: DC-jack + / - before the first power-up.
Sources: control/wiring.md - Marlin pins_RAMPS.h, TMC2209 datasheet rev 1.09,
BTT TMC2209 V1.3 manual + schematic, RAMPS 1.4 KiCad netlist.

Driver jumpers and UART addresses (2)

Socket	Role	MS1	MS2	MS3	Address
X	A (top-left)	off	off	off	0
Y	B (top-right)	ON	off	off	1
Z	C (bottom)	off	ON	off	2
E0	key	ON	ON	off	3

Never fit an MS3 jumper. Socket E1 stays empty.

Motor currents, set over UART (7)

Driver	Run	Hold	Why
Dials A, B, C	1.0 A RMS	0.5 A	until the dial torque is measured
Key	0.6 A RMS	0.3 A	then 2 x the measured minimum, <= 1.0 A
Firmware ceiling	1.2 A	-	BTT: active cooling above 1.2 A

Parts you fit (5, 10) - full list: docs/bom.md

Where	Ref	Part	Status
Pigtail	R1	1 kOhm in line in the TX2 lead (the only UART resistor)	have
Pigtail	-	6 F-F jumpers, one end cut off each; heat shrink	ordered
PSU	-	DC jack 5.5 x 2.1 mm -> screw terminal, -> RAMPS 5A	ordered
PSU	-	20 AWG red + black, adapter -> RAMPS 5A terminal	ordered
RAMPS	S1	7 mm push button (start / stop)	have
Wiring	-	F-F jumpers, ~20 x 10-20 cm (6 of them for the pigtail)	ordered
Key	-	the motor's own ~1 m cable -> E0 motor header	have
Pigtail	-	small zip tie: the splice to the deck	ordered

Revision 2026-10-08.1 removed the remote driver board, the 6-conductor cable + Phoenix connectors, the hub 12 V branch + fuse, and C1a/C1b; 2026-10-08.2 replaced the hub board (perfboard) with the UART pigtail. RAMPS F1, F2, D1, the six 100 uF and the 10 k pull-ups are on the RAMPS.

+12 V +5 V (logic, VIO) GND UART bus
STEP / DIR / EN, button DIAG (stall) motor coils jumper / tie / spare
bought module board you build already on the RAMPS / Mega: nothing to buy

Fichet safe robot — wiring schematic **Sheet 7 of 7**

Pin map, jumpers, currents, parts

2026-10-08 · revision 2026-10-17.1 · from control/wiring.md